MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP
(HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Value Addition Course (VAC) - MG3VACCFS200 - E-WASTE MANAGEMENT AND RECYCLING - SCHEME
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Part A
Short Answer Type Questions
Answer any 10 questions
Each question carries 1 marks

1. Evaluate pure metals and metal oxides [E] / [CO4]

Precious metals: Au, Ag, Pt, Pd recovered from e-waste by hydrometallurgy/electrometallurgy .
Metal oxides (Cu_2O, Fe_2O_3, ZnO, etc.) are obtained via electrodeposition or oxidation.
Both forms serve as valuable raw materials for nanomaterials and electronic components.

2. Biological method to enhance phytoremediation [An] / [CO2]

Use of mycorrhizal fungi and soil microorganisms improves metal uptake and plant biomass .

3. Differentiate electrowinning and electrorefining [A] / [CO3]

Electrowinning: extraction of metal from leach solution onto cathode.
Electrorefining: purification of impure metal anode; pure metal is deposited on cathode .

4. Discuss about electrodeposition [U] / [CO1]

Electrochemical process depositing metal ions onto electrode surface. Applied in recovery of Cu, Te, and rare earths; rate improved using ionic liquids .

5. Potential concern related to phytovolatilization [E] / [CO4]

Volatilized contaminants (Hg, Se, As) may re-enter the environment via precipitation; not fully removed .

6. Classify contaminants (table form) [U] / [CO1]

Heavy metals	Pb, Cd, Hg, Cu, Ni, Zn
Organic pollutants	PCBs, PBDEs, PAHs
Gaseous/by-products	Dioxins, furans

7. Contaminants phytoremediation can address [U] / [CO1]

Trace metals (Pb, Cd, Cu, Ni, Zn, As, Hg, Se).
Organic compounds (PCBs, hydrocarbons) .

8. Process to selectively precipitate metals in hydrometallurgy[A] / [CO3]

Chemical precipitation by pH control or double-salt formation after leaching

9. Process converting solid into soluble ions [A] / [CO3]

Leaching: dissolution of metals using acids (H_2SO_4, HCl, HNO_3) .

10. Nanomaterials used to improve phytoremediation [A] / [CO3]

Metal/metal-oxide NPs (Fe_3O_4, ZnO, TiO_2, Ag) enhance uptake, enzyme activity, and stress tolerance .

11. Innovative way to reduce e-waste [U] / [CO1]

Circular economy: urban mining, eco-design, and biorecycling using microorganisms .

12. How genetic engineering improves phytoremediation [An] / [CO2]

Transgenic plants with genes for metal transporters, chelators, or detox enzymes show higher uptake .

Part B
Short Essay Type Questions
Answer any 4 questions
Each question carries 5 marks

13. Disadvantages and challenges of electrokinetic remediation [E] / [CO4]

Slow process for low-conductivity soils.
Possible pH gradient, electrode corrosion, and high energy demand .

14. Write oxidation and reduction reactions [A] / [CO3]

Any correct answer - 5 marks
example:
Oxidation: $Fe^{2+} \rightarrow Fe^{3+} + e^-$
Reduction: $Cu^{2+} + 2e^- \rightarrow Cu(s)$

15. Classify WEEE into five categories [U] / [CO1]

(i) Large household appliances
(ii) Small appliances
(iii) IT & telecom equipment

(iv) Consumer electronics
(v) Lighting, tools, toys, medical devices .
Give an example for each category

16. Compare landfill and biological method [An] / [CO2]

Landfill → cheap, causes leachate pollution.
Biological (phytoremediation/bioremediation) → eco-friendly, slower .

17. Limitations of rhizofiltration [U] / [CO1]

Effective only in water; requires constant maintenance; limited root depth .

18. Five metals in printed circuit boards and importance [U] / [CO1]

Any correct answers - 5 marks
Examples:Cu (conductivity), Au (corrosion resistance), Ag (soldering), Pb (solder), Ni (plating) .

Part C
Essay Type Questions
Answer any 2 questions
Each question carries 10 marks

19. Adsorption and desorption processes in e-waste [U] / [CO1]

Adsorption: binds metal ions on surfaces (e.g., activated carbon, biochar).
Desorption :regenerates adsorbent.
Used for Cu, Pb, Ni recovery from leachates .

20. Improving phytodegradation [A] / [CO3]

Use of plants producing high enzyme levels (peroxidase, dehalogenase).
Combine with microbes, surfactants, or electric fields .

21. Applications and limitations of electrokinetic remediation [E] / [CO4]

Removes metals and organics from soils.
Can be combined with PRB or phytoremediation .
Limitations: high cost, electrode degradation, uneven field distribution